

AI DECEPTION ANALYZER (CLI-BASED)

Fundamentals of AI and ML – Course Project



Submitted By

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1. TITLE

AI Deception Analyzer: A CLI-Based Machine Learning System for Detecting Deceptive Text and Emotional Tone



2. INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) have enabled systems to understand and analyze human language. One such application is detecting deception in textual communication.

The AI Deception Analyzer is a **Command Line Interface (CLI)-based application** that analyzes text input and predicts whether it is truthful or deceptive. It also identifies the emotional tone of the text.

This project combines **Natural Language Processing (NLP)** techniques with machine learning models to create a lightweight and efficient system.



3. MOTIVATION OF THE PROJECT

In everyday communication, identifying deception is difficult. People often hide truth using complex or emotional language.

Motivation behind this project:

- To explore **real-world applications of NLP**
 - To build a **practical AI system**
 - To understand how language patterns indicate deception
 - To develop a **CLI-based application** as per requirement
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4. PROBLEM STATEMENT

Detecting deception in human communication is a challenging problem due to:

- Lack of clear indicators
- Subjective interpretation of language
- Variations in communication styles

Problem:

Design a system that can analyze text input and predict whether it is deceptive or truthful while also identifying emotional tone.



5. OBJECTIVE OF THE PROJECT

- Build a CLI-based AI application
- Apply NLP techniques to text data
- Train a machine learning model for classification

- Detect emotional tone using rule-based logic
 - Store and retrieve analysis history
 - Ensure usability via terminal commands
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6. EXISTING METHODS

Existing deception detection methods include:

1. **Human Judgment**
 - Based on intuition and experience
 2. **Psychological Analysis**
 - Uses behavioral cues
 3. **Advanced AI Models**
 - Deep learning models like BERT
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7. PROS AND CONS OF EXISTING METHODS

Pros:

- Advanced models provide high accuracy
- Psychological methods consider context

Cons:

- Human judgment is unreliable
 - Deep learning models are complex and resource-heavy
 - Not easily deployable in simple environments
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8. HARDWARE AND SOFTWARE REQUIREMENTS

◆ **Hardware:**

- Basic computer/laptop
- Minimum 4GB RAM

◆ **Software:**

- Python 3.8+
 - VS Code / Terminal
 - Libraries:
 - scikit-learn
 - pandas
 - numpy
 - joblib
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9. METHODOLOGY AND GOAL

◆ **Methodology:**

1. Data collection (sample dataset)
2. Text preprocessing
3. Feature extraction using TF-IDF
4. Model training using Logistic Regression
5. Emotion detection using keyword rules
6. CLI interface for user interaction

Goal:

To create a **functional, lightweight, and interpretable AI system** for deception detection.

10. FUNCTIONAL MODULES DESIGN AND ANALYSIS

1. Input Module

- Accepts user input via CLI

2. Processing Module

- Converts text into numerical features

3. Prediction Module

- Uses ML model to predict deception

4. Emotion Module

- Detects emotional tone

5. Storage Module

- Saves results in JSON file

6. CLI Module

- Handles commands and user interaction
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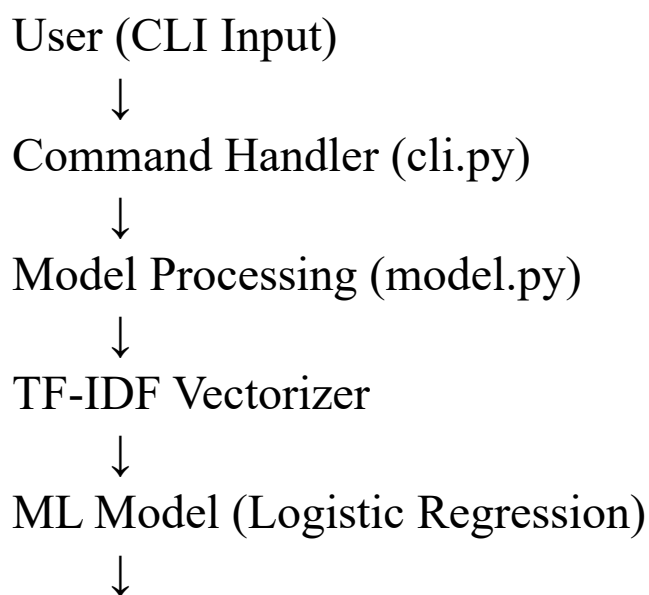
11. ALGORITHM DEVELOPMENT

Step-by-Step Algorithm:

1. Start program
2. Display CLI menu

3. Accept user command
 4. If command = analyze:
 - Take input text
 - Transform using TF-IDF
 - Predict using model
 - Detect emotion
 - Display result
 - Save to history
 5. If command = history:
 - Load stored data
 - Display results
 6. If command = exit:
 - Terminate program
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12. SOFTWARE ARCHITECTURAL DIAGRAM



Emotion Detection (utils.py)



Output Display



JSON Storage (history.json)



13. CODING

The project is implemented using:

- Python for backend logic
- scikit-learn for machine learning
- Modular structure (cli, model, utils)

Key files:

- cli.py → CLI logic
 - model.py → ML prediction
 - train.py → training
 - utils.py → emotion detection
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14. OUTPUT

Example Output:

Enter command: analyze

Enter text: I swear I didn't do anything wrong

Truth Probability : 0.30

Deception Probability : 0.70

Emotion : Neutral

15. KEY IMPLEMENTATIONS

- TF-IDF Vectorization
- Logistic Regression Model
- CLI-based interaction system
- JSON-based data persistence

16. SIGNIFICANT PROJECT OUTCOMES

- Successfully implemented AI-based CLI system
- Demonstrated NLP and ML concepts
- Built a complete working pipeline
- Improved understanding of real-world AI applications

17. TESTING AND REFINEMENT

- Tested with multiple text inputs
- Handled empty and invalid inputs
- Improved CLI usability
- Verified model predictions

18. REAL-WORLD APPLICATIONS

- Fraud detection
- Chat analysis

- Customer feedback systems
 - Online moderation tools
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19. CONTRIBUTION / FINDINGS

- Simple models can still provide useful insights
 - Language patterns play a key role in deception
 - CLI applications are efficient and practical
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20. LIMITATIONS

- Small dataset → limited accuracy
 - Emotion detection is basic
 - Cannot fully understand human psychology
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21. CONCLUSIONS

The AI Deception Analyzer demonstrates how machine learning can be used to analyze human language and detect patterns of deception.

Although limited in accuracy, the system provides a strong foundation for future improvements and real-world applications.

22. FUTURE ENHANCEMENTS

- Use deep learning models (BERT)

- Improve dataset size
 - Add voice input
 - Add visualization tools
 - Deploy on cloud
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23. REFERENCES

- scikit-learn documentation
 - Python documentation
 - NLP research articles
 - Online ML resources
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FINAL NOTE

This project successfully fulfills the requirements of:

- AI-based system ✓
- CLI-based application ✓
- Real-world problem ✓
- Structured implementation ✓